MIL-G-49032A(ER)
16 APRIL 1979

SUPERSEDING
MIL-G-49032(EL)
13 MAY 1974

MILITARY SPECIFICATION

GENERATOR, INTERFERENCE SG-886( )-T/UR

This specification is approved for use by the U.S. Army Communications and Electronics Materiel Readiness Command (USACERCOM), also U.S. Army Electronics Research and Development Command (USAERADCOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers the requirements for the construction and testing of one type of electronic counter - countermeasures (ECCM) training device designated Generator, Interference SG-886( )-T/UR, hereinafter called "generator". The generator is used to noise or tone modulate standard Army voice communication sets, such as AN/VRC-12, AN/PRC-25, AN/PRC-77 and AN/GRC-106 with pre-selected output levels.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, ERADCOM, Fort Monmouth, N.J. 07703, ATTN: DELET-DP, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6940

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS.

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Sheet and Strip, Thin, Gauge, Polyolefin
NN-P-71	Pallet, Material-Handling, Wood, Stringer Construction 2-Way and 4-Way(Partial)
QQ-S-781	Strapping, Steel, and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation Packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

STANDARDS, MILITARY (Cont'd)

MIL-STD-147	Palletized Unit Loads for 40" x 48" Pallets
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461A	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests - Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Equipment

DRAWINGS

USAERADCOM

DLSM-B-867168	Interference Generator, GS-886( )-T/UR Data List
SM-A-867216	Wide Band Amplifier Array
SM-A-867226	Microcircuit, Selected

(Copies of specifications, standards, and drawings required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS.

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

3.2 Construction. The generator shall be constructed in accordance with the requirements of this specification and the drawings listed in the USAERADCOM Data List DLSM-B-867168.

3.2.1 Wide band amplifier array. The wide band amplifier array SM-A-867216 shall be selected in accordance with procedure SM-A-867226 prior to use.

3.3 Parts, materials and process; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.3.1 Finish. The generator shall be finished in accordance with MIL-F-14072 and the drawings listed on DLSM-B-867168.

3.4 Marking.

3.4.1 General. Marking shall conform to MIL-M-13231 (see 4.4). The front panel marking shall be Group I as described in that specification.

3.4.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.3 Serial number. Each generator shall have a serial number.

3.5 Performance.

3.5.1 General characteristics. The generator, powered by one BA-1363/U battery is a device which modulates tactical FM and AM/SSB communication sets with noise and tone, as well as with signals externally fed through from an auxiliary receiver. The signals are fed to the set's audio terminals by means of cable assemblies with compatible audio connectors. The device consists of a generator unit assembly secured in its own equipment case. Attached to the generator unit are two cable assemblies which extend from their storage area in the equipment case. Two adapter cable assemblies and a spare battery clip are included in the equipment case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

3.5.2 Current drain. The generator shall use one BA-1363/U battery (12.15 Vdc) with a rated life of 250 milliampere hours. Current drain in Mode 1 shall be no greater than 3 milliamperes at 12 volts and in Mode 2 shall be no greater than 5 milliamperes at 12 volts (see 4.7.1).

3.5.3 Battery voltage monitor. The monitor shall indicate when the battery voltage is too low for satisfactory operation of the generator. The monitor is a meter which has a scale marked in green and red to indicate the proper voltage for operation. When the monitor indicates red in Mode 1 or 2, the battery voltage shall be less than 9 volts. However, when in the green, the voltage shall be greater than 9 volts (see 4.7.2).

3.5.4 Audio output time and noise. The generator shall have two types of audio outputs, audio tone and audio random noise.

3.5.4.1 Tone frequency. The audio tone frequency for each of the two modes shall be as specified below: (see 4.7.3.1 and 4.7.3.2).

Mode	Frequency
1	980 Hz ± 100 Hz
2	860 Hz ± 100 Hz

3.5.4.2 Tone and noise level outputs. The generator shall produce variable voltage outputs for each tone and noise mode from zero volts to the level specified below: (see 4.7.3.3).

Function	Mode	Voltage (RMS)
Tone	1	7 millivolts (mv) to 40 mv
Tone	2	15 mv to 90 mv
Noise	1	4 mv to 57 mv
Noise	2	9 mv to 87 mv

Also, the ratio of noise to spectral content shall be 3 db or greater.

3.5.5. Input and output impedance.

3.5.5.1 Input impedance. The input impedance of the generator at the receiver cable assembly (marked "FROM RCVR") U-229/U connector shall be 1.0 K ohms ± 10% at a frequency of 1 kHz. (see 4.7.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

3.5.5.2 Output impedance. The output impedance of the generator at the transmitter cable assembly (marked "TO XMTR") U-229/U connector shall be 39 ohms ± 10% at a frequency of 1 kHz. The output impedance of the generator at the adapter cable assembly (marked "TO XMTR") U-229/U connector, when connected, shall be 500 ohms ± 10% at a frequency of 1 kHz (see 4.7.5).

3.5.6 Recorder, external modulation. When the generator is used with a commercial recorder, the generator shall provide external modulation with combination of tone or noise at various output levels to a US Army standard transmitter (AN/PDC-77) (see 4.7.6).

3.5.7 Compatibility with voice communication sets. When the generator is used with US army standard communication sets (transmitter and receiver), various levels of tone or noise interference shall be generated (see 4.7.7).

3.6 Environmental. The generator, shall meet the following service conditions without degradation in specified performance of 3.5.

3.6.1 Temperature.

3.6.1.1 Low temperature. The generator shall be operable without degradation in specified performance at ambient temperature as low as -40 degrees F and withstand exposure to ambient air temperature as low as -70 degrees F (see 4.9.1.1).

3.6.1.2 High temperature. The generator shall be operable without degradation in specified performance at ambient temperatures as high as +131 degrees F and withstand exposure to ambient air temperature as high as +160 degrees F (see 4.9.1.2).

3.6.2 Altitude. The generator shall be operable without degradation in specified performance at altitude up to 10,000 feet above sea level and withstand air transportation at altitude up to 17,500 feet above sea level (see 4.9.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

3.6.3 Humidity. The generator shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to relative humidities as high as 100 percent at all ambient air temperatures to +85 degrees F and relative humidities corresponding to a dew point of +85 degrees F at all temperatures between +85 degrees F and +120 degrees F, and low relative humidity of 5 percent at +120 degrees F (see 4.9.3).

3.6.4 Sand and dust. The generator shall withstand, in both operating and nonoperating conditions, exposure to fine sand and dust particles with wind speeds of up to 20 knots. After testing as specified in 4.9.4, the equipment shall meet specified performance requirements.

3.6.5 Salt fog. In both operating and nonoperating conditions, the generator shall withstand the salt fog test of 4.9.4. Upon completion of test, the generator, when examined visually with the aid of a 10 power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surfaces; or exfoliation of metal (see 6.3.6). There shall be no clogging or binding of controls or other moving parts after subjecting the generator to the 48 hour salt fog exposure period or after subjection to the subsequent 48 hour drying period (see 4.9.5).

3.6.6 Rain. The generator shall be operable without degradation in specified performance and shall sustain no physical damage or leakage of water during periods of heavy precipitation with intermittent wind (see 4.9.6).

3.6.7 Fungus. After the fungus test of 4.9.7, the generator shall be visually examined, using a 10 power magnifier. The generator shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.3.3) as evidenced by growth colonization (see 6.3.2), which includes branching (see 6.3.1) and sporulation (see 6.3.4) on or within each cubic foot, or fraction thereof, of generator assembly volume. Isolated instances of partical tubular germination (see 6.3.5) shall not be included in this evaluation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

3.6.8 Vibration and shock. The generator shall suffer no degradation, deterioration, or change in tolerance limits which could in any manner affect operational or service maintenance requirements when subjected to vibrations encountered during field transportation by military vehicle and when subjected to shocks encountered in rough handling and servicing (see 4.9.8, 4.9.9, 4.9.10 and 4.9.11).

3.7 Electromagnetic interference (EMI). The generator shall comply with the following emission and susceptibility requirements of Notice 4, MIL-STD-461A: (see 4.9.12)

RE02	14 kHz-1000 MHz Broadband
RE02.1	14 kHz-1000 MHz Narrowband
RS02	Magnetic Induction Field Spike
RS03	10 kHz-200 MHz Electric Field

3.8 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.8).

3.9 Reliability. The generator shall demonstrate a lower test MTBF(θ_1) of 1000 hours and an upper test MTBF(θ_0) of 2000 hours when operated under the reliability test conditions stated herein (see 4.7.9).

3.10 Burn-in. Each generator on order shall meet the requirements specified in 3.5, without subsequent processing, after subjection to burn-in as specified in 4.6.

3.11 Workmanship. The generator shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Flux and cleaning agent for soldering
Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Self-tapping screws

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

3.11 Workmanship (cont'd)

b. In this specification:

3.3.1 Finish

3.12 Systems safety engineering.

3.12.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.10).

3.12.2 Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm) and exposed corners to a minimum of 0.05 inch (1.3 mm).

3.12.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.13 Human factors engineering. Design configuration as affecting operation, maintainability and handling shall be maintained through compliance of configuration criteria to the requirements of MIL-STD-1472.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging) (see 4.3).

b. Inspection covered by subsidiary documents. (see 4.4).

c. Quality conformance inspections:

(1) Quality conformance inspection of equipment before packaging. (see 4.5).

(2) Quality conformance inspection of packaging. (see 4.11)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish nine first article units of the complete generator.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in Table I and shall be performed in the order specified herein.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification.

Item	Where Required
Parts, materials and processes	3.3
Finish, protective	3.3.1
Marking	3.4

4.5 Quality conformance inspection of equipment before packaging. Each unit of each lot of generators shall be burned-in prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

Notes	Inspections	Req Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5-9
1/	1. Inspection covered by subsidiary documents	See 4.4	4.4	Inspection to be performed on all units.				
1/	2. Group A inspection	See Table II		Inspections to be performed on all units.				
1/	3. Group B inspection	See Table III		Inspections to be performed on all units.				
1/	4. Burn-in	3.10	4.6	Inspections to be performed on all units.				
2/	5. EMI	3.7	4.9.12	1				
2/	6. ENVIRONMENTAL							
	Low temperature	3.6.1.1	4.9.1.1	2				
	High temperature	3.6.1.2	4.9.1.2	3				
	Altitude	3.6.2	4.9.2	4				
	Humidity	3.6.3	4.9.3	5				
	Sand and dust	3.6.4	4.9.4			1		
	Salt fog	3.6.5	4.9.5			2		
	Rain	3.6.6	4.9.6		3			
	Fungus	3.6.7	4.9.7		2			
	Vibration	3.6.8	4.9.8		1			
	Bounce, loose cargo	3.6.8	4.9.9	6				
	Shock, drop	3.6.8	4.9.10				2	
	Bench handling	3.6.8	4.9.11				1	
2/	7. Group D reliability	3.9	4.7.8					1

NOTES: 1/ The inspections 1 through 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

2/ The number in the unit columns in the table covered by inspections 5, 6, 7 specify the order of inspections for the indicated unit.

3/ The equipment shall be thoroughly washed, cleaned, dried, and refurbished after the fungus and salt fog tests before proceeding with subsequent inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the general inspection levels and AQLs indicated in Table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection level of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans for the AQL's listed in Table III shall be S-4. All defects shall be considered major.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order satisfactory to the Government.

TABLE II. Group A inspection.

Inspection	Req. Para.	Insp. Para.	AQL Major	ALQ Minor
Visual and mechanical	3.11	4.10	1.0%	4.0%
Electrical				
Current drain	3.5.2	4.7.1	1.0%	1/
Battery voltage monitor	3.5.3	4.7.2	1.0%	1/
Tone frequency	3.5.4.1	4.7.3.1 4.7.3.2	1.0%	1/
Tone and noise level outputs	3.5.4.2	4.7.3.3	1.0%	1/
Input impedance	3.5.5.1	4.7.4	1.0%	1/
Output impedance	3.5.5.2	4.7.5	1.0%	1/

1/ All electrical defects are considered major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

TABLE III. Group B inspection

Inspection	Req. Para.	Test Para.	AQL	
			Major	Minor
Recorder, external modulation	3.5.6	4.7.6	4.0%	1/
Compatibility	3.5.7	4.7.7	4.0%	1/
Interchangeability	3.8	4.8	4.0%	1/

1/ All defects are considered major.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1. The inspection shall consist of the inspection specified in Table IV.

4.5.3.1 Sampling for Group C inspection. Four samples of the complete equipment shall be selected at random, 1 sample for each Subgroup I through and including Subgroup IV in Table IV. The first samples shall be selected at the start of production from the first quality conformance inspection lot. These samples shall constitute the Group C requirements for the first 25 units produced. Thereafter, 4 samples of the complete equipment shall be selected at random, 1 sample for each Subgroup I through IV in Table IV from every 75 units or fraction thereof greater than one-half. If the fraction is less than or equal to one-half, the fraction shall be considered part of the previous Group C lot for sampling purposes. Salt fog and fungus shall not be repeated after the first production lot Group C inspection.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order acceptable to the Government.

4.5.3.3 Group C failure. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

TABLE IV. Group C inspection

Inspection	Req. Para.	Test Para.
Subgroup I		
Altitude	3.6.2	4.9.2
Humidity	3.6.3	4.9.3
Rain	3.6.6	4.9.6
Subgroup II		
Low temperature	3.6.1.1	4.9.1.1
High temperature	3.6.1.2	4.9.1.2
Vibration	3.6.8	4.9.8
Subgroup III		
Bounce, loose cargo	3.6.8	4.9.9
Shock, drop	3.6.8	4.9.10
Bench handling	3.6.8	4.9.11
Subgroup IV		
EMI	3.7	4.9.12
Sand and dust 1/	3.6.4	4.9.4
Salt fog 1/	3.6.5	4.9.5
Fungus 1/	3.6.7	4.9.7

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units from lots which have been subjected to and passed Group A and Group B inspection.

TABLE V. Group D inspection

Inspection	Req. Para.	Test Para.
Reliability	3.9	4.7.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.5.5.1 Sampling for Group D inspection. Five (5) generators shall be selected for reliability qualification test from the first article units in accordance with Table I. Five (5) generators shall be selected at random for the production reliability test from every 200 equipments or fraction thereof greater than one-half. If the fraction is less than or equal to one-half, the fraction shall be considered part of the previous Group D lot for sampling purposes.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.6 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Burn-in. Each deliverable generator shall be subjected to a 48 hour burn-in period, conducted under the standard ambient conditions of 3.1, MIL-STD-810. This period shall include 24 hours of full operation in Mode 1 and 24 hours of full operation in Mode 2. The sequence of operation shall be in accordance with Table VI.

4.6.1 Monitoring of burn-in. As a minimum the following performance parameters shall be monitored during the burn-in period:

Test	Paragraph
Audio Output Levels	3.5.4
Current Drain	3.5.2
Input/Output ac Impedance	3.5.5

4.6.2 Failures. Failures will be allowed during the first 12 hours of operation in Mode 1 and the first 12 hours of operation in Mode 2, however, the final 12 hours of operation in each mode must be failure free. If a failure does occur in the final 12 hours of operation in either mode, the equipment shall be repaired and returned to burn-in until it survives the 12 hour failure free period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.6.3 Compliance. Units passing burn-in shall be submitted to the electrical measurements of Group A and B inspection without benefit or repair, cleaning, realignment or other changes, unless otherwise specified herein. If after successful completion of burn-in, the generator fails either of the electrical measurements in Group A or Group B inspection, the generator shall be reworked and resubmitted to the last 12 hours of the burn-in period and shall successfully complete the 12 hour failure free period of burn-in prior to being resubjected to Groups A and B inspection.

4.6.4 Conditions preceding burn-in. A thermal survey and vibration survey shall be conducted in accordance with MIL-STD-781. The time required to stabilize at the low temperature shall determine the length of the temperature cycle to be conducted during burn-in.

4.6.5 Test conditions. Each generator shall be subjected to the following test conditions during burn-in:

a.	Temperature	$73^{o} \pm 18$ F
b.	Relative humidity	$50\% \pm 30\%$
c.	Vibration frequency	Any selected frequency within the range 20 Hz to 30 Hz (excluding resonant points).
d.	Vibration amplitude	$\pm$ 1.5 g's

The generator shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 48 hour period of operation. The direction of vibration shall be vertical to the normal mounting plane for 5 minutes and then lateral to that plane for 5 minutes. Where it is not feasible to vibrate the generator in these 2 directions, the vertical direction shall be used. After the 10 minute vibration period the generator shall be operated at each of the switch settings for the length of time as indicated in Table VI. Should a failure occur, the generator shall be repaired and the burn-in test shall be started over, except that the 10 minute vibration period need not be repeated when it is certain that the failure was not a result of vibration. Should repetitive failures occur, corrective action shall be taken to eliminate the defect from the generator. A record shall be kept of all failures. The 48 hour period of operation specified above may be composed of successive 8 hour periods to conform with standard working hours provided that the generator is operated at the generator settings and periods specified in Table VI (see 3.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

TABLE VI. Burn-in

Noise-Off-Tone Switch Setting	Failures	Period of Operation
Tone	1 Allowed	12 hours
Tone	2 Not allowed	12 hours
Noise	1 Allowed	12 hours
Noise	2 Not allowed	12 hours

4.7 Performance.

4.7.1 Current drain. The current drain of the battery in the generator shall be measured with the generator set to TONE, MODE 1 and again with the generator set to NOISE, MODE 1. In neither instance shall the current drain exceed 3 milliamperes at 12 Vdc. The current drain shall then be measured with the generator set to TONE, MODE 2 and again with the generator set to NOISE, MODE 2. In neither instance shall the current drain exceed 5 milliamperes at 12 Vdc (see 3.5.2).

4.7.2 Battery voltage monitor. A dc power supply capable of supplying any voltage from zero to 12 Vdc shall be connected to the generator in lieu of the BA-1363/U battery. The generator shall be turned on and the MODE switch shall be set at MODE 1. The supply voltage shall be varied from zero to 12 Vdc. The MODE switch shall then be set to MODE 2 and the supply shall again be varied from zero to 12 Vdc. The battery voltage monitor shall meet the requirements of 3.5.3 for both MODE 1 and MODE 2 operation (see 3.5.3).

4.7.3 Audio outputs, tone and noise.

4.7.3.1 Tone frequency, MODE 1. The frequency of the audio tone output shall be measured with the generator set to TONE, MODE 1. The measured frequency shall meet the requirements of 3.5.4.1.

4.7.3.2 Tone frequency, MODE 2. The frequency of the audio tone output shall be measured with the generator set to TONE, MODE 2. The measured frequency shall meet the requirements of 3.5.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.7.3.3 Tone and noise level outputs. The variable voltage outputs for each of the tone modes and for each of the noise modes shall be measured and shall meet the requirements of 3.5.4.2.

4.7.4 Input impedance. The impedance shall be measured at the receiver cable assembly U-229/U connector with the generator turned on and a 1 kHz signal applied. The input impedance shall meet the requirements of 3.5.5.1.

4.7.5 Output impedance. The impedance shall be measured at the transmitter cable assembly U-229/U connector with the generator turned on and a 1 kHz signal applied. Also, the impedance shall be measured at the transmitter adapter cable assembly (attached to the transmitter cable assembly) connector with the generator turned on and a 1 kHz signal applied. The output impedance shall meet the requirements of 3.5.5.2.

4.7.6 Recorder, external modulation. When the generator is connected with a commercial recorder in accordance with FIGURE 1 and setting the tone or noise switch in their appropriate position (tone or noise) and varying the output control, the generator shall meet the requirements of 3.5.6. The commercial recorder shall be adapted to the generator jack A3J1.

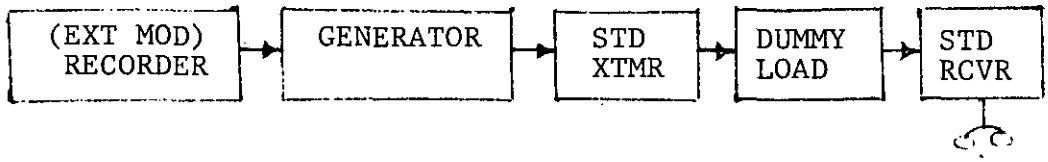


FIGURE 1. Block Diagram of the Interference Simulation Set-Up With External Modulation (Recorder).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.7.7 Compatibility with voice communication sets. The generator shall be used with U.S. Army standard communication sets in accordance with FIGURE 2. With the tone or noise switch in their appropriate position and varying the output control, the generator shall meet the requirements of 3.5.7.



FIGURE 2. Block Diagram of the Interference Simulation Set-Up.

4.7.8 Reliability. The contractor shall conduct a reliability acceptance test in accordance with Test Plan III C of MIL-STD-781. Five (5) generators will be selected by the Government from the first article test models and the specified production rate, stated in the contract to undergo reliability testing (see 3.9).

4.7.8.1 Test conditions.

Temperature	$50^{o}+5^{o}-0^{o}C$ ($122^{o}F$ to $131^{o}F$)
Temperature Cycling	None
Vibration	1.5g's $\pm$ 10% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points on the generator. The duration of vibration shall be at least 10 minutes during each hour of generator operating time.
Equipment on/off cycling	Turn on and let temperature stabilize, hold for 3 hours, then turn off and let temperature stabilize. This cycle shall continue throughout the test.
Duty cycle	50% MODE 1 and 50% MODE 2.
Input voltage	Nominal specified voltage 12.0 Vdc.
Input voltage cycling	None

4.7.8.2 Failure definitions. For the purpose of reliability testing, the following failure definition shall apply: a failure is any inability of the equipment to perform its required functions. This includes any event which causes departure from performance as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.7.8.3 Performance measurements. As a minimum, the following parameters shall be measured and recorded at least every 48 hours during the test.

Parameter	Requirement paragraph
Current drain	3.5.2
MODE 1 TONE	3.5.4.1
MODE 2 TONE	3.5.4.1
Audio outputs	3.5.4.2
Input/output impedance	3.5.5

4.8 Interchangeability.

4.8.1 Inspection for dimensional interchangeability. The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, keying pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major units.

e. Size and form of special threads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.8.2 Electrical interchangeability. All like modules and replaceable parts shall be physically and functionally interchanged without modification of such items and the performance tests of paragraph 4.7 shall be performed to determine compliance with 3.8. However, electrical retuning and realignment of modules during production assembly operations is permissable.

4.9 Environmental. The generator shall be tested in accordance with the methods and procedures of MIL-STD-810 given below. The limits of allowable performance degradation shall be as specified in 3.5 of this specification, and shall form the criteria for passing or failing each environmental test unless stated otherwise. Successful completion of any particular test shall be considered evidence of meeting the corresponding environmental condition.

4.9.1 Temperature.

4.9.1.1 Low temperature. The generator shall be subjected to the test of Method 502.1, Procedure 1, of MIL-STD-810. The storage temperature shall be -70 degrees F and the operating temperature shall be -5 degrees F. After stabilization of the test item in Step 4, a fresh battery may be inserted for operation of the generator. Prior to testing, the generator shall meet full specification performance as specified in 3.5.4 and 3.5.5.

4.9.1.2 High temperature. The generator shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. The operating temperature shall be +131 degrees F. Prior to testing, the generator shall meet full specification performance as specified in 3.5.4 and 3.5.5.

4.9.2 Altitude. The generator shall be subjected to the test of Method 500.1 of MIL-STD-810, except Step 2 shall be 17,500 feet. Prior to testing, the generator shall meet full specification performance as specified in 3.5.4 and 3.5.5.

4.9.3 Humidity. The generator shall be subjected to the test of Method 507.1, Procedure II of MIL-STD-810. In Step 6, measurements shall be taken during the first 5 hours of each cycle. Prior to testing the generator shall meet full specification performance for the measurements specified in 3.5.4 and 3.5.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.9.4 Sand and dust. The generator shall be tested in accordance with Method 510.1, Procedure 1 of MIL-STD-810. Prior to testing, the generator shall meet full specification performance for the measurements specified in 3.5.4 and 3.5.5 (see 3.6.4).

4.9.5 Salt fog. The generator shall be tested in accordance with Method 509.1, Procedure 1 of MIL-STD-810. Operation of the generator shall not be required, and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the test items surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface.

4.9.6 Rain. The generator shall be subjected to the test of Method 506.1, Procedure 1 of MIL-STD-810. The generator shall be operated during the last 10 minutes of the 30 minute period and shall meet the requirements of 3.5.4. Prior to testing, the generator shall meet full specification performance for the measurements specified in 3.5.4 and 3.5.5. (see 3.6.6).

4.9.7 Fungus. The generator shall be subjected to the fungus test specified in Method 508.1, Procedure 1 of MIL-STD-810. There shall be abundant growth colonization (6.3.2) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the generator is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the generator. Inability of the generator to meet the requirements of 3.6.7 shall constitute failure of this test.

4.9.8 Vibration. The generator shall be subjected to the vibration test of MIL-STD-810, Method 514.2, Procedure X. Prior to testing, the generator shall meet the full specification performance for the requirements of 3.5.4 and 3.5.5 (see 3.6.8).

4.9.9 Bounce, loose cargo. The generator shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810. Prior to testing, the generator shall meet full specification performance as specified in 3.5.4 and 3.5.5 (see 3.6.8).

4.9.10 Shock, drop. The generator shall be tested in accordance with Method 516.2, Procedure II of MIL-STD-810. Prior to testing, the generator shall meet the full specification performance as specified in 3.5.4 and 3.5.5 (see 3.6.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

4.9.11 Bench handling. The generator shall be tested in accordance with Method 516.2, Procedure V of MIL-STD-810. Prior to testing, the generator shall meet full specification performance as specified in 3.5.4 and 3.5.5. (see 3.6.8).

4.9.12 EMI. The generator shall be tested for compliance with the requirements of 3.7. The test shall be in accordance with Notice 4, MIL-STD-461A, as implemented by a contractor prepared, government approved EMI test plan. (see 3.7)

4.10 Visual and mechanical inspection. The generator shall be examined for the defects listed in MIL-STD-252. An inspection shall also be performed to verify compliance with those portions of 3.12 (Systems Safety Engineering) which can be determined visually. (see 3.11).

4.11 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of 5 herein. The rough handling test will be performed only when specified (see 6.2).

4.12 Human engineering verification. Maintenance of design configuration to the requirements of 3.13 shall be verified by visual and mechanical inspection.

5. PACKAGING.

5.1 Preservation. Preservation shall be Level A, B or Commercial, as specified (see 6.2.f).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Generator, Interference SG-886( )-T/UR shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Generator, Interference SG-886( )-T/UR shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. Not required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be individually unit packed Method 1C-1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

5.1.1.4.2 Generator, Interference SG-886( )-T/UR. Each Generator, Interference SG-886( )-T/UR shall be individually unit packed Method III as follows: Coil cable assemblies to an appropriate diameter and secure them in the cable storage compartment of the equipment case. Close and secure case cover. The secured case shall be cushioned on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be inclosed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2.f). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Generator, Interference SG-886( )-T/UR, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.2.2 Level B. A quantity of Generator, Interference SG-886( )-T/UR, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape, banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of Generator, Interference SG-886( )-T/UR, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES.

6.1 Intended use. Generator, Interference SG-886()-T/UR is designed for use as an electronic counter-countermeasures (ECCM) training device that will noise or tone modulate Army voice communication sets. This training device can convert such equipments to communication jammers with the noise or tone modulation provided from the device. Internal power is supplied from one Battery BA-1363/U.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Marking and shipping of equipment.

c. Place of final inspection.

d. Technical literature required.

e. Radio Set AN/PRC-77 and Battery BA-1363/U shall be Government furnished property (loaned) for use in SG-886( )-T/UR tests.

f. Level A, Level B or Commercial preservation and packing (see Section 5).

g. When rough handling test is required (see Section 4.11).

6.3 Definitions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

6.3.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.3.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.4 Sporulation. Sporulation is the formulation of minute unicellular reproductive or dormant bodies, called spores.

6.3.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.3.6 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Custodian:
Army ER

Preparing Activity
Army ER

Proj No. 6940-A069

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander, ERADCOM
ATTN: DELET-RP
Fort Monmouth, NJ 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-G-49032A(ER) GENERATOR, INTERFERENCE SG-886( )-T/UR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:44Z
Check the source to verify that this is the current version before use.